

T₉ Consider the Lotka-Volterra Predator-Prey model

$$\frac{dx}{dt} = ax - pxy \quad (\text{prey equation})$$

$$\frac{dy}{dt} = -by + qxy \quad (\text{predator equation})$$

with the parameters $a, b, p, q > 0$.

- (a) Describe how the system of differential equations models a predator-prey interaction.
- (b) Find the equilibrium solutions. Write a tweet (x-eet) for each of the equilibria describing them in a real world context).
- (c) Calculate the Jacobian and decide on the stability of each equilibrium solution.
- (d) One of the equilibrium solutions is stable, but not asymptotically stable / attracting (solutions starting nearby, don't converge towards the equilibrium, but also don't diverge away).

Let us show this fact:

- i. To show this, use the system of ODEs above to find an expression for $\frac{dy}{dx}$.
 - ii. Solve the differential equation you obtained for $y(x)$.
 - iii. Your solution has the form $E(x, y) = C$ for an arbitrary constant C .
 - iv. Conclude that the solution of the system of ODEs, “live” on the level sets of a function, which are closed curves. So the system has periodic orbits.
 - v. Use python to visualize the level sets of E .
 - vi. Simulate a solution using Runge-Kutta's method (or a built-in python function). What do you observe?
- (e) The Lotka-Volterra model has provided insight into real-world ecological phenomenon. Consider the example:

A beach-side community became panicked after a string of shark attacks one summer. They decided to aggressively hunt the sharks, which dramatically reduced their numbers near the beach. However, the following year there was a considerable increase in the number of shark sightings.

Recalling that the sharks are predators of fish, explain the situation.

Use a sketch to describe what probably happened in your level set graph.

T₁₀¹ Now let us consider a modified Lotka-Volterra model to account for logistic growth of the prey population:

$$\begin{aligned}\frac{dx}{dt} &= ax \left(1 - \frac{x}{K}\right) - pxy && \text{(logistic prey equation)} \\ \frac{dy}{dt} &= -by + qxy && \text{(predator equation)}\end{aligned}$$

with $a = b = p = q = 1$ and $K > 0$.

- (a) What value of K recovers the original Lotka-Volterra predator-prey model?
- (b) Find the equilibrium solutions and the values of K for which they are all valid. How would you label them for a lay person?
- (c) Classify the stability of each equilibrium solution.
- (d) Sketch a phase portrait and describe the effects of including a logistic prey growth in the Lotka-Volterra model.

¹Based on the 2024 test 1.